Supplementary material

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Supplement to "A benchmark of cell-type annotation methods for single-cell data". Figures and tables referenced from the main text as Supplementary Figure S1–S4 and Supplementary Table S1–S3.

1 Supplementary figures

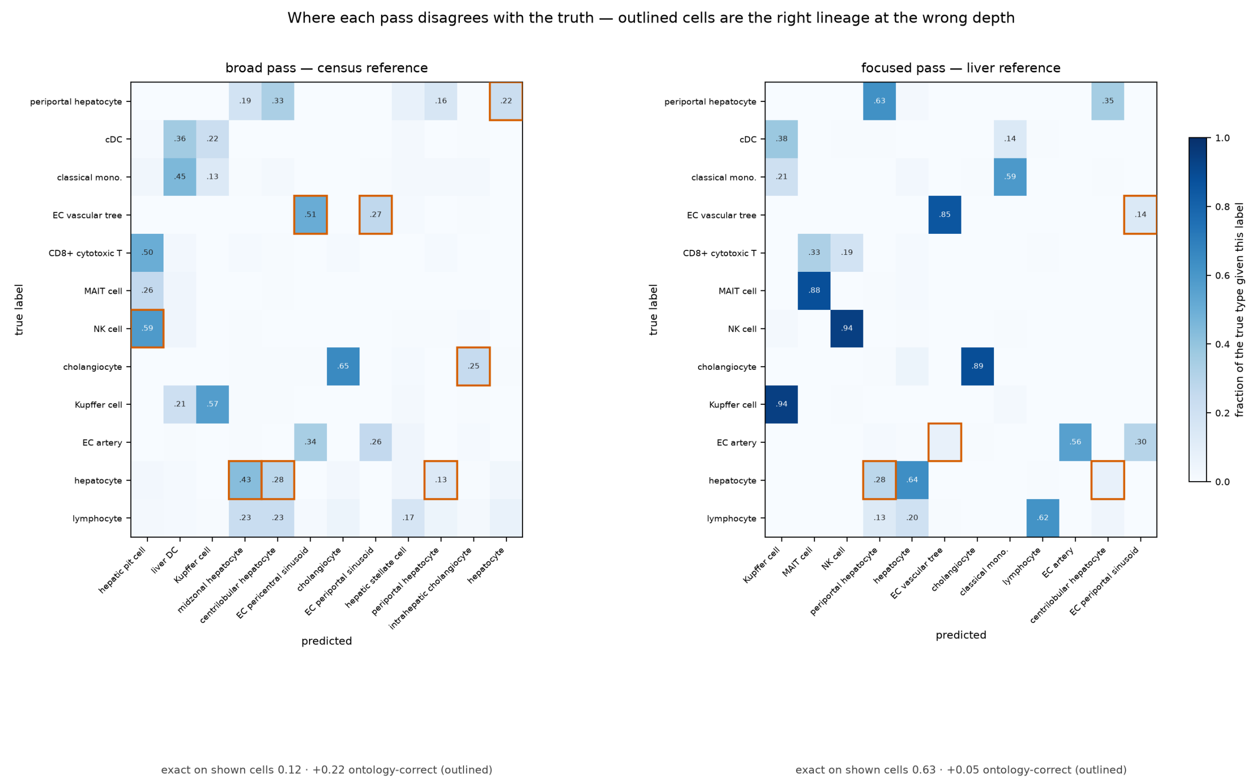

Figure S1. Confusion matrices for the broad and focused passes on the withheld cross-study liver query. Each cell is the fraction of a true type receiving that label; outlined cells are off-diagonal calls that are an ancestor or descendant of the truth in the Cell Ontology, i.e. error under exact match and credit under ontology-aware concordance. The scores below each panel are means over the twelve truth types drawn, not over the whole query, so they run lower than the query-wide figures quoted in the main text. The broad pass scores 0.12 exact and recovers +0.22 through the ontology — its mistakes are the right lineage at the wrong depth (hepatocyte → midzonal or centrilobular hepatocyte; NK cell → hepatic pit cell, the Cell Ontology term for a liver NK cell).

The focused pass scores 0.63 exact, +0.05: it is already right most of the time, so the ontology has little left to recover.

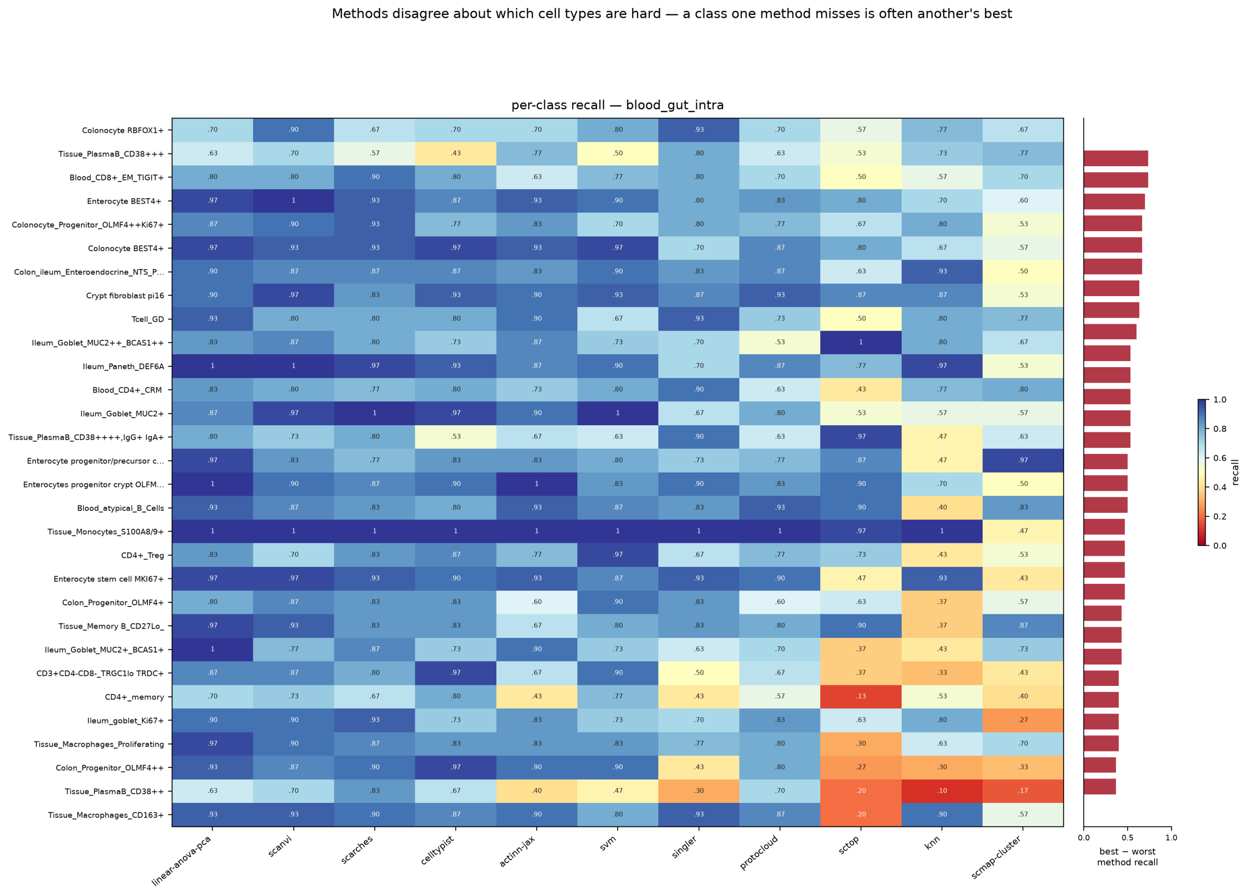

Figure S2. Per-class recall for eleven methods on the 86-type blood+gut split, ordered by how much the methods disagree (best minus worst recall, right-hand strip). The best and worst method differ by a median of 0.30 recall per class and up to 0.73; mean pairwise Spearman between methods' per-class recall is 0.58. Class size is not the explanation, and on this split it cannot be: capping per label leaves **84 of the 86 classes holding exactly 30 test cells**, with the remaining two at 29 and 14. There is no size variation left for rarity to act through, so the disagreement is about which types a method finds hard, not how many examples it saw.

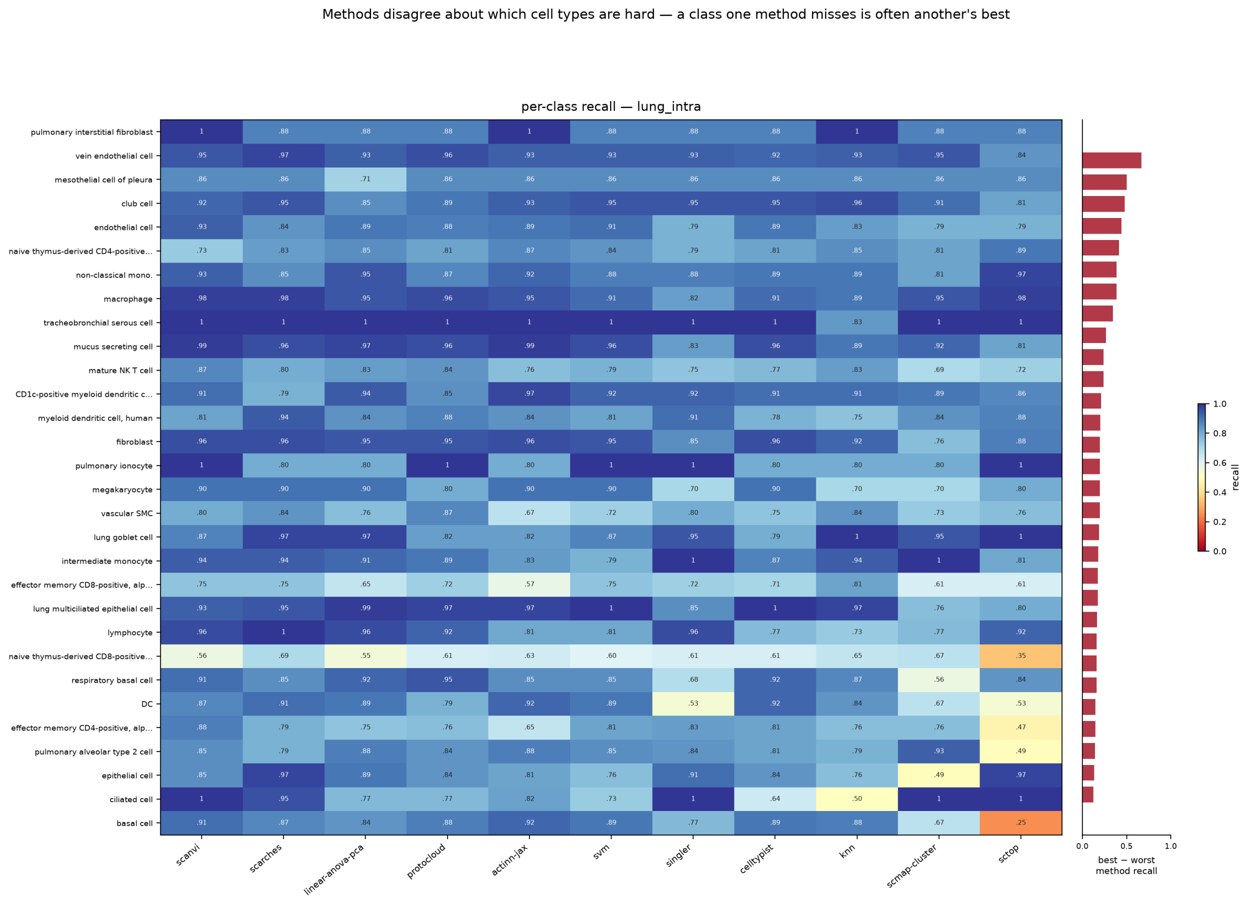

Figure S3. Per-class recall on the 46-type lung split, drawn as in Figure S2. Methods agree more here than on blood+gut: mean pairwise Spearman 0.65, median best-minus-worst 0.16.

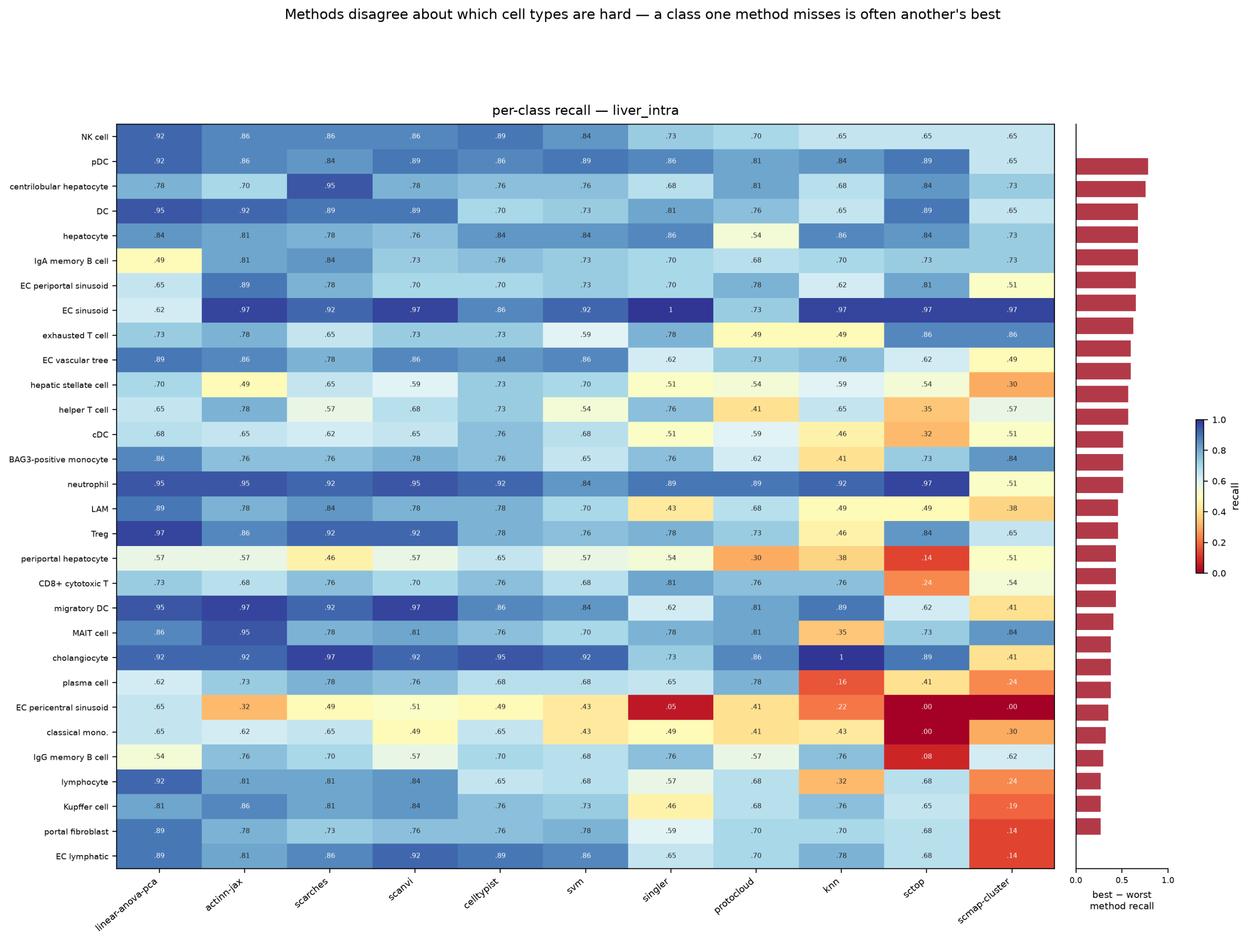

Figure S4. Per-class recall on the 36-type liver split, drawn as in Figure S2. Mean pairwise Spearman is 0.66, but the median best-minus-worst gap is 0.43 — the widest disagreement of the three splits, and a reminder that a high rank correlation between methods still leaves large per-class differences.

2 Supplementary tables

method	source	tier	engine	rejection	runtime
actinn-jax	[Ma & Pellegrini 2020]	classical	JAX MLP	confidence threshold	JAX
SVM	[Pedregosa 2011]	classical	linear SVM (SGD)	—	scikit-learn
kNN	[Pedregosa 2011]	classical	k-nearest neighbors	—	scikit-learn
CellTypist	[Domínguez Conde 2022]	linear	L2 logistic regression	prob threshold	scikit-learn
linear-anova-pca	[Pedregosa 2011] †	linear	normalize → ANOVA → PCA(220) → logreg	prob	scikit-learn

method	source	tier	engine	rejection	runtime
scTOP	[Yampolskaya 2023]	parameter-free	rank z-score class-average projection	—	NumPy
SingleR	[Aran 2019]	correlation	Spearman + fine-tuning	—	R
scmap-cluster	[Kiselev 2018]	correlation	centroid cosine	yes (unassigned)	R
scANVI	[Xu 2021]	deep	scVI semi-supervised VAE	prob	scVI
scArches	[Lotfollahi 2022]	deep	scANVI reference surgery	prob	scVI
ProtoCloud	[Guo & Ding 2026]	deep	prototype VAE + LRP attribution	ambiguity flag	PyTorch
scPRINT	[Kalfon 2025]	foundation	pretrained transformer, zero-shot	—	PyTorch
Pan-human Azimuth	[Sarkar et al. 2026]	pretrained	8-level hierarchical NN, fixed 382-leaf typology	trained `Unassigned`	TensorFlow

Table S1. The benchmarked methods: model family (*tier*), the engine each one runs, whether it can decline to call a cell (*rejection*), and the runtime stack it was run on. Bold marks the methods added to this panel here.

† `linear-anova-pca` has no method paper: it is a baseline assembled here from scikit-learn components, tuned deliberately to be the strongest simple competitor we could build.

dataset	source	split	tissue	#types	genes	notes
lung_intra	[Travaglini 2020]	within-dataset	lung (Krasnow)	46	Ensembl	300 cells/type ref
lung_cross	[Sikkema 2023] → [Travaglini 2020]	cross-dataset	lung (HLCA → Krasnow)	46	Ensembl	different lab/protocol
liver_intra	[Edgar 2026]	within-dataset	liver (HLiCA)	36	Ensembl	150 cells/type
liver_cross	[Edgar 2026]	cross-**study**	liver (HLiCA)	34	Ensembl	train 6 studies → test withheld study
blood_gut_intra	[Alegbe 2026]	within-dataset	blood + gut (IBDverse)	86	Ensembl	high cardinality; no CL ids

dataset	source	split	tissue	#types	genes	notes
pbmc	[10x Genomics 2016]	within-dataset	PBMC (pbmc3k)	8	symbols	small-n; scPRINT skips (symbols)

Table S2. The six benchmark datasets. *split* separates within-dataset holdouts from cross-dataset and cross-study transfer; *genes* records whether the matrix is keyed by Ensembl ids or gene symbols.

dataset	actinn-jax	best method (acc)
lung_intra (46 types)	0.894	ProtoCloud 0.932
lung_cross (cross-dataset)†	0.358 exact / 0.752 ontology	ProtoCloud 0.374 / **0.791 ontology**
liver_intra (36 types)	0.802	linear-anova-pca 0.804
liver_cross (cross-study)	**0.686 exact / 0.731 ontology**	actinn-jax
blood_gut (86 types)	0.860	linear-anova-pca 0.902
pbmc (8 types)	0.913	scArches 0.940

Table S3. Per-dataset accuracy: actinn-jax against whichever method leads that dataset. † on lung_cross the exact-match score is a vocabulary artifact, so ontology concordance is the meaningful column.